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Thermodynamic properties of aqueous Ge(IV) hydroxide complexes from 25 to 350°C: Implications for the behavior of germanium and the Ge/Si ratio in hydrothermal fluids

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ABSTRACT—The stoichiometry and thermodynamic properties of Ge(IV) hydroxide complexes were generated from both solubility and potentiometric measurements. The solubility of the tetrahedral germanium oxide (GeO_2(tetr)) was measured at temperatures from 25 to 350°C in acid to alkaline solutions at the saturated vapor pressure of the system (P_{sat}). Potentiometric measurements were performed on GeO_2-KOH aqueous solutions at temperatures from 21 to 200°C and P_{sat} using a pH solid-contact glass electrode. Results indicate that $Ge(OH)_4^\circ$(aq) is the dominant Ge-bearing species at concentrations up to at least 0.05 m over a wide range of pH (0–8) and temperatures (20–350°C). $GeO(OH)_3^-$ forms in significant amounts only in alkaline solutions (pH > 8–9). These results were combined with the available low-temperature solubility data on the hexagonal germanium oxide (GeO_2(hex)) and the thermodynamic properties of GeO_2(tetr) and GeO_2(hex) to generate $Ge(OH)_4^\circ$(aq) and $GeO(OH)_3^-$ thermodynamic parameters within the framework of the revised HKF equation of state (Helgeson et al., 1981; Tanger and Helgeson, 1988). Calculations carried out using these parameters indicate that the distribution of Ge hydroxide species as a function of pH and temperature is similar to that of silicon hydroxide complexes. However, the significant differences between $Ge(OH)_4^\circ$(aq) and $Si(OH)_4^\circ$(aq) enthalpies of formation and heat capacities can lead to large variations with temperature of Ge/Si ratios in solutions in equilibrium with Ge-bearing silicates. For example, calculations show that the Ge/Si ratio in a fluid in equilibrium with a Ge-bearing wollastonite ($Ca(Si,Ge)O_3$) increases by an order of magnitude when temperature is raised from 25 to 500°C. This can be responsible for the high values of Ge/Si ratios measured in high temperature crustal fluids. *Copyright © 1998 Elsevier Science Ltd*

1. INTRODUCTION

Germanium is a trace component in the Earth's crust and natural waters. However, the wide use of Ge in the manufacture of electronics, infra-red intruments, and optic fibres makes it an element of high technological value (Geldron, 1983). Germanium can be also used as a tracer for certain geochemical processes such as continental weathering and oceanic hydrothermal activity since the relative intensity of these processes can be related to Ge/Si ratios in rivers and ocean waters (Froelich et al., 1985a; Mortlock and Froelich, 1987; Murnane and Stallard, 1990; Floelich et al., 1992; Mortlock et al., 1993). Modeling of Ge behavior in natural systems as well as prospecting and treatment of Ge-bearing ores requires knowledge of the thermodynamic properties of its aqueous species and Ge-bearing solid phases. The purpose of this paper is to determine the thermodynamic properties of Ge(IV) hydroxide complexes which dominate Ge speciation in organic ligand-free natural fluids and to use these data for the interpretation of the Ge/Si ratios observed in hydrothermal fluids.

The chemical properties and inorganic geochemical pathways of Ge and Si are very similar (Cotton and Wilkinson, 1966; Froelich et al., 1985a,b). Germanium substitutes for Si in mineral lattices (Goldschmidt, 1958), but it exhibits distinctly lithophile, siderophile, chalcophile, and organophile properties depending on the chemical environment (Bernstein, 1985). Various mineral associations such as pegmatites, low temperature sulfides, iron oxides and hydroxides, and coals are very enriched in Ge (Bernstein, 1985, Pokrovski, 1996). In rivers that are distant from industrial regions and sources of coal combustion, the value of Ge/Si atomic ratio is about $0.6 \cdot 10^{-6}$ that is somewhat lower than that of most crustal rocks ($1–2 \cdot 10^{-6}$; Martlock and Froelich, 1987; Froelich et al., 1992). This can be related to the preferential uptake of Ge vs. Si by weathering products such as clays and iron oxihydroxides, therefore reflecting the relative intensity of chemical weathering (Murnane and Stallard, 1990). However, the Ge/Si ratio increases significantly in waters enriched in organic matter (to $\sim 10^{-5}$ in tropical watersheds; Viers et al., 1997). In the oceans, the biochemistry and distribution of inorganic Ge and Si are very similar. Inorganic Ge behaves as a heavy stable isotope tracer for Si in biogenic opal (Froelich et al., 1992; Mortlock et al., 1993). Besides inorganic germanium, high concentrations of monomethyl- and dimethylgermanium have been detected in ocean waters and estuaries. These species dominate Ge speciation in seawater accounting for > 70% of the total Ge (Froelich et al., 1985a; Lewis et al., 1985, 1988). In hydrothermal fluids, the atomic Ge/Si ratio is much higher than that of rivers, oceans, and crustal rocks attaining values of $8–14 \cdot 10^{-6}$ in hot springs in mid-ocean ridges (Froelich et al., 1993), 10^{-5} in Icelandic hydrothermal systems (Arnorsson, 1984), and even $10^{-4}–10^{-3}$ in hot springs of the Massif Central, France (Criaud and Fouillac, 1986). Both Ge concentration and Ge/Si ratios increase with temperature in most hydrothermal waters. The absence of positive correlations between Ge concentrations and those of important inorganic ligands like Cl^-, HCO_3^-, H_2S, Na^+ suggests germanium does not form complexes with these ligands (Koga, 1967; Goleva and Vorob'yeva, 1967; Arnorsson, 1984). Available experimental data on the stability of Ge-fluoride complexes (Ryss and Kulich, 1965; Naumov et al, 1974; Ciavatta et al., 1990) imply

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that these species can be important only in acid (pH < 3) fluor-rich (>0.01m) solutions that are very rarely found in hydrothermal environments. The redox potential of most hydrothermal and surficial environments is too high to cause significant formation of divalent Ge species such as Ge^{2+}, $GeOH^{+}$ or $Ge(OH)_2^{\circ}$ (Baes and Mesmer, 1976; Arnorson, 1984). As a result, tetravalent Ge (like Si) is present as hydroxide complexes in most natural fluids with the exception of seawater and organic-rich surficial waters.

The thermodynamic properties of silicon hydroxide species, $Si(OH)_4^{\circ}$ and $SiO(OH)_3^{-}$, are accurately known to high temperatures (Baes and Mesmer, 1976; Walther and Helgeson, 1977; Busey and Mesmer, 1977). In contrast, very few data are available on their germanium analogs, $Ge(OH)_4^{\circ}$(aq) and $GeO(OH)_3^{-}$, at both low and high temperatures. The present study was initiated to provide a coherent set of thermodynamic properties for Ge hydroxide complexes which can allow a better characterization of Ge behavior in natural systems. For this purpose, the solubility of germanium oxide (GeO_2(tetr)) was determined at temperatures up to 350°C in acid to basic solutions (pH = 1.5–10). The pH of $Ge(OH)_4$-KOH aqueous solutions was measured from 20 to 200°C using a high temperature glass electrode. These results allow the calculation of the thermodynamic properties of $Ge(OH)_4^{\circ}$(aq) and $GeO(OH)_3^{-}$ to high temperatures and permit a comparitive modeling of Ge and Si behavior in natural hydrothermal fluids.

2. MATERIALS AND METHODS

2.1. Synthesis of GeO_2(tetr)

Tetragonal Ge(IV) dioxide (GeO_2(tetr)), which is a structural analog of rutile, was synthesized from commercial hexagonal GeO_2(hex) (Aldrich 99.999%) following the method of Faktor and Carasso (1965). The allotropic transformation of GeO_2(hex) to GeO_2(tetr) occurs at high temperature in the presence of a catalyst. The catalyst used in this study was lithium germanate (Li_2GeO_3) prepared from equimolar quantities of germanium hexagonal dioxide (GeO_2(hex)) and lithium carbonate by slowly heating the mixture in a platinum crucible to 1250°C. After attainment of 1250°C, this mixture was held at this temperature for 2 h. The melt was cooled and ground to a fine powder in an agate mortar. GeO_2(hex) was mixed with 0.5 wt% of lithium germanate and heated in a covered platinum crucible at 960°C for 12–48 h. X-ray powder analysis showed that the transformation was complete after 12 h and the synthesized product was pure well-crystallised tetragonal Ge(IV) dioxide (JCPDS card 9-379). From 10 to 15 grams of this powder were placed with 250 mL of MilliQ water in a rocking titanium autoclave and heated to 250°C. After 3 weeks, the solution and the solid phase were separated while maintaining the temperature at 250°C to avoid possible precipitation of a poor-crystallised oxide. This procedure was repeated two to three times. After this treatment, the powder was rinsed repeatedly with MilliQ water and oven dried at 60°C for 3 days. The specific surface area of this solid was 0.5 ± 0.05 m^2/g as measured by the BET method using Kr as the adsorbing gas. Thermogravimetric analysis showed a single endothermic peak at 1030°C which corresponds to the tetragonal-hexagonal transition of the oxide. Infra-red and X-ray Photoelecton (XPS) spectra indicated the only presence of GeO_2(tetr) bands and Ge/O atomic ratio of 1:2 at the surface of the oxide, respectively. SEM analysis showed the resulting solid to be composed of 1 μm crystals associated in agglomerates of about 20–100 μm.

2.2. Analytical Methods

Aqueous Ge concentrations were measured by three different methods according to the concentration of the solution: flame atomic absorption (AAS), ICP-MS, and colorimetry. Germanium standard solutions were prepared by weight dilution of an Aldrich 1005 ppm Ge standard solution with deionized water. Solutions containing from 5 to 100 ppm of Ge were analysed by flame atomic absorption using acetylene-nitrogen protoxide flame and an absorption line of 261.5 nm. Reproducibility of this method was better than 5%, and the detection limit defined as the 2σ standard deviation was found to be 0.5 ppm. Analyses of standard solutions with concentrations of the reagents used in the experiments (NaCl, NaAc, $B(OH)_3$, HCl, NaOH) to 0.1m showed no influence on the calibration curve. For Ge concentrations ranging from 5 to 100 ppb, ICP-MS analyses with ^{73}Ge and ^{74}Ge isotopes were performed. To diminish matrix effects, standard and blank solutions had the same chemical compositions as the experimental solutions. Reproducibility and detection limit were 3% and 0.05 ppb, respectively. Solutions with intermediate Ge concentrations between 1 and 20 ppm were analysed colorimetrically by the molybdate blue method using ascorbic acid as reducing agent. The analyses were performed on an Autoanalyser-II Technicon according to the molybdate blue method for silica analysis (Koroleff, 1976). The reproducibility and detection limit of Ge analyses were 5% and 0.1 ppm, respectively. Germanium concentrations of the same experimental solutions analysed by the three methods were within the same limit of 5%.

2.3. Solubility Measurements

GeO_2(tetr) solubility was measured at temperatures from 25 to 350°C at saturated vapor pressure and over the pH range 1.5–10. Different reactor vessels were used over different temperature ranges. At 25°C, experiments were conducted with high solid/solution ratios (R > 1:10) in 5 or 10 cm^3 plastic syringes capped and placed in a thermostated bath. The syringe samples were shaken periodically by hand. To monitor attainment of equilibrium, 1–3 mL of solution were taken periodically and filtered through a 0.05μm Millipore filter. At 50 and 90°C, solubility measurements were performed in 10–250 cm^3 polypropylene or teflon flasks immersed in water thermostated baths. The use of reactors having different volumes permitted the variation of solid/solution ratio (R) from 1:2 to 1:1000 without increasing the mass of solid in the reactor. Samples containing 5–10 mL of solution were taken using a syringe and rapidly (<1 min) filtered through a 0.05 μm Millipore filter. At temperatures from 150 to 350°C, experiments with solid/solution ratios varying from 1:30 to 1:1000 were conducted in ~380 cm^3 titanium autoclaves placed in temperature controlled (±2°C) rocking furnaces. These reactors allow injection of solids and sampling of the solution at the run temperature. Details on the vessel design and experimental procedure are given elsewhere (Castet, 1991; Castet et al., 1993; Benezeth, 1994; Pokrovski, 1996). Solution aliquots were periodically removed from the reactor through a 2 μm titanium filter inserted in the reactor cover and re-filtered at room temperature with a 0.05 μm Millipore filter. At temperatures from 250 to 350°C, experiments with high solid/solution ratios (1:30 ≤ R ≤ 1:3) were performed in 35 cm^3 Ti-alloy (Ti, Mo, Al, VT-8) autoclaves placed in temperature controled (±2°C) ovens. The solution was filtered after quenching the reactor in cold water. For the same experimental conditions, the variation of quenching time from 5 to 60 min did not change the Ge concentration in the solution. No difference in the Ge concentrations was observed after filtering the same solutions through Millipore filters of different porosity ranging from 0.01 to 0.45 μm in the investigated temperature range of 25–350°C. After selected dissolution runs at each temperature, blank experiments were performed in the same reactors with Ge-free HCl-NaOH solutions. Aqueous Ge concentratons after these experiments were always below 0.1 ppm, suggesting that Ge adsorption on the reactors wall was negligeable at all temperatures.

To cover the wide range of investigated pH values, solubility measurements were performed in deionized water and $HClO_4$, HCl-NaCl, NaAc-HAc, $B(OH)_3$-NaOH, and NaOH-NaCl aqueous solutions prepared from corresponding Prolabo Normapur Reagents. The ionic strength of these solutions varied from 0 to 0.1 m. The pH of the initial and sampled solutions was measured at 25°C using a commercial combination glass electrode (see below). pH values at the run temperatures were calculated using the extended Debye-Hückel equation for the activity coefficients, and the dissociation constants for HCl°, NaCl°, NaOH°, acetic acid, boric acid, and H_2O reported by Ruaya and Seward

(1987), Johnson et al. (1992), Castet et al. (1993), Mesmer et al. (1989), Mesmer et al. (1972), and Sweeton et al. (1974), respectively.

2.4. Potentiometric Measurements

Potentiometric measurements on KOH-H_4GeO_4 aqueous solutions from 20 to 90°C were performed using a commercial combination pH glass electrode (SCHOTT H61) with a Philips-PW9422 digital pH-meter. The electrode was fixed in the plastic cover of a ~60 cm^3 Pyrex, water thermostated (±0.2°C), potentiometric cell equipped with a magnetic Teflon-coated stirring bar. The electrode calibration was performed using 0.1 m HCl ($pH_{25°C}$ = 1.10) and 0.01 m HCl ($pH_{25°C}$ = 2.04) solutions, and the NBS/DIN phthalate ($pH_{25°C}$ = 4.008), phosphate ($pH_{25°C}$ = 6.865), and borate ($pH_{25°C}$ = 9.180) buffer solutions. Excellent linear relationships between the electrode potential and solution pH were obtained from 20 to 90°C with the electrode slopes corresponding to the theoretical Nernst slopes in the limit of 2%.

Measurements at temperatures from 50 to 200°C were performed using a high-temperature pH-selective glass electrode created by the St.-Petersburg company Potential (Beliustin et al., 1992). Details on the electrode and potentiometric cell construction are given in Pokrovski et al. (1993, 1995, 1996b). This electrode has an electrochemically reversible solid contact with the glass ion-conducting membrane and contains no internal solution. The inner surface of the Li-Al-borosilicate glass selective to H^+ is covered by a thin Li-Sn alloy that provides a reversible contact and remarkably stable potential. The upper operating temperature is 200°C because the melting point of the Li-Sn alloy is ~215 ± 10°C. At 25°C, the electrode resistance is too high (>1000 MΩ) to make precise measurements, however, this resistance decreases rapidly with increasing temperature and allows electrical potential measurements in the 50–200°C temperature range with a precision of 0.1mV using an ordinary potentiometer. In the present study, a Tacussel-2000 pH-meter with an input impedance $10^{13}\Omega$ was used. The reference electrode is an external AgCl/Ag, 3.2 M KCl electrode with a construction analogous to that used by Krukov et al. (1966) and improved by Pokrovski (1996) and Pokrovski et al. (1995,1996b). It consists of a Teflon tube containing in its upper part the Ag/AgCl element and filled with a 3.2 M KCl reference solution and glass fiber to diminish possible thermal convection of the reference solution between the upper and the lower part of the electrode. The Ag/AgCl element is located outside the reaction cell and kept at 25°C by thermostated water circulation. On the lower part on the tube situated in the reaction cell is fixed another Teflon tube opened in its upper part and contaning the same 3.2M reference KCl solution. The level of the KCl solution in this tube is 1–2 cm higher than that of the test solution. In the lower extremity of the tube is incorporated a 0.1 μm Teflon filter which serves as a liquid junction between the reference and the test solution. This construction permits a slow outflow of the KCl reference solution to the test solution and thus diminish significantly the liquid junction potential. Experiments were carried out in a ~350 cm^3 titanium autoclave equiped with the reference electrode and 2–3 pH glass electrodes. This reactor was placed in an oven and kept at a constant (±0.3°C) temperature. The solution was continuously stirred with a magnetic Teflon-coated stirring bar.

Electrode calibration was performed over the temperature range 50–200°C at saturated vapor pressure using 0.1m HCl, 0.01m HCl + 0.015m KCl, 0.025 m KCH_3COO + 0.025m CH_3COOH, 0.05m NH_3 + 0.025m NH_4Cl, 0.05m $B(OH)_3$ + 0.025m KOH, and 0.01m KOH + 0.015m KCl solutions. The pH of these solutions was calculated using the extended Debye-Hückel equation for the activity coefficients and the dissociation constants for HCl°, H_2O, acetic acid, boric acid and ammonia reported by Ruaya and Seward (1987), Sweeton et al. (1974), Mesmer et al. (1989), Mesmer et al. (1972), and Fisher and Barnes (1972), respectively. The uncertainty on the calculated pH values taking account of those on the dissociation constants and activity coefficients is estimated to be ±0.02 pH units up to 200°C. The liquid junction potentials were considered constant over the pH range investigated. The electrode calibration curves are linear with slopes corresponding to the Nernst values in the limit of ±2% at all temperatures. These calibration curves allow determination of experimental solution pH values with a precision better than ±0.05 pH units.

Two experimental solutions containing 0.0184 m GeO_2 + 0.010 m KOH, and 0.008 m GeO_2 + 0.005 m KOH were prepared with MilliQ water by dissolving GeO_2(hex) powder (Aldrich, 99.999%) in KOH solutions obtained by dilution of a 1M KOH standard solution (Merck, Titrisol) and then filtered through a 0.45 μm Millipore filter. The experimental solutions were placed in the potentiometric cell and heated to the desired temperature. The e.m.f. of the cell stabilized within 1–2 min. of the attainment of the temperature of interest and remained constant (within 1 mV) for at least 0.5 h. Temperature was then raised to another desired value, and a new potential was recorded. To check the reversibility of the electrode response, measurements were also performed on cooling. The corresponding potentials were the same within 5–10 mV. To increase the reproducibility of measurements and minimize the effect of possible changes in pH with time, only potentials recorded on heating were used for pH determinations. The pH's of experimental solutions were obtained from the measured e.m.f. values using the calibration curve parameters at the corresponding temperature. Estimated uncertainties in pH measurements obtained from 2 to 3 electrodes were in the range of ±0.05 pH units.

3. RESULTS AND DISCUSSION

3.1. Solubility Measurements

3.1.1. Kinetics and Equilibrium Criteria

Preliminary experiments at temperatures from 25 to 150°C performed with GeO_2(tetr) powder prepared as explained above showed that the higher the solid/solution ratio, the higher Ge aqueous concentration measured at steady-state. Moreover, in experiments performed at higher temperatures (200 and 250°C) Ge concentrations measured as a function of time increased to a maximum value and then decreased which demonstrates recrystallisation of the GeO_2 powder. The dependence of aqueous Ge steady-state concentration on the solid/solution ratio at low temperatures and its decrease as a function of time at high temperature both indicate that the solid used in this study contained ultra fine particles and/or crystalline surface defects.

In order to remove ultra fine particles and defects, successive dissolution runs of the same GeO_2(tetr) powder were conducted at temperatures from 25 to 150°C. Each investigated powder was subjected to dissolution until aqueous Ge concentration attained a steady-state. Then, the powder was recovered, washed with deionized water, and used in a new dissolution run at the same conditions (pH, T, and R). This procedure was repeated until Ge concentrations measured at steady-state in two successive dissolution runs were the same. An example of succesive dissolutions of a GeO_2 powder at 150°C is shown in Fig. 1. Dissolution experiments carried out at higher temperatures (T > 150°C) with such a pretreated powder yield steady-state Ge aqueous concentrations that are independent of the solid/solution ratio, although a few runs pass through a maximum (Fig. 2). It was thus assumed that true equilibrium with GeO_2(tetr) was achieved in these runs. Times required to reach steady-state range from a week at T ≤ 250°C with R > 1:30 to several months at T ≤ 90°C and R < 1:20.

Several experiments were also performed at 200°C starting from solutions supersaturated with respect to GeO_2(tetr) to demonstrate equilibrium reversibility. Supersaturation was achieved by heating a solution with GeO_2(tetr) powder (R ~ 1:40) at near neutral pH (5–7) to 300°C in a closed titanium reactor. After equilibrium was reached, the temperature was lowered to 200°C. However, even after two months of reaction, Ge concentrations remained several times higher than those

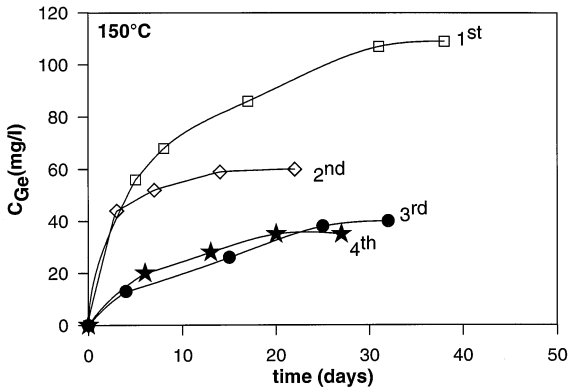


Fig. 1. Germanium concentrations measured at 150°C as a function of time in four successive GeO_2(tetr) dissolution runs in neutral ($4 < pH < 6$) solutions with solid/solution ratios from 1:50 to 1:30. The 3rd and 4th runs correspond to the runs N and O, respectively, listed in Table 1. It can be seen that Ge steady-state concentration values measured in the 3rd and 4th runs are approximately the same.

attained in undersaturation runs. Furthermore, measurements of the dissolution/precipitation rates of GeO_2(tetr) at 250°C and near-neutral pH using a mixed-flow kinetic reactor yielded very low precipitation rates ($<5 \cdot 10^{-10}$ mol m^{-2} s^{-1}) even in solutions supersaturated by a factor of five with respect to GeO_2(tetr). This can explain how Ge solutions can remain for a long time supersaturated with respect to GeO_2(tetr). It should be noted that very low quartz precipitation rates have been also measured close to equilibrium (saturation index ~ 2) in weakly acidic solutions at 200°C (Gautier et al., 1996). Consequently, only GeO_2(tetr) solubilities measured from undersaturation were considered in this study.

3.1.2. *Results of GeO_2(tetr) solubility experiments and comparison with previous data*

The results of the solubility measurements performed with the pretreated GeO_2(tetr) powder are presented in Table 1

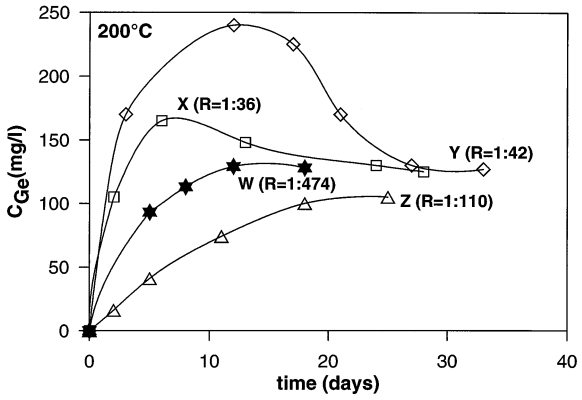


Fig. 2. Ge concentrations measured at 200°C as a function of time in neutral solutions in the dissolution runs of pretreated GeO_2(tetr) (see text) at different solid/solution ratios (R). The letters refer to the runs reported in Table 1. It can be seen that all the runs yield approximately the same steady-state aqueous Ge concentrations.

together with reacting solution compositions and pH values measured at 25°C and calculated at the run temperature. It can be seen in Table 1 that GeO_2(tetr) solubilities are independent of pH in acid and neutral solutions ($1 \leq pH \leq 8$), ionic strength, and oxygen fugacity. This is consistent with the formation of the neutral germanium hydroxide species (germanic acid) up to at least 350°C according to the reaction:

$$GeO_2(tetr) + 2H_2O = Ge(OH)_4^\circ(aq) \qquad (K_{tetr}) \qquad (1)$$

The predominance of neutral $Ge(OH)_4^\circ(aq)$ species in acid and neutral solutions at low temperature is also supported by the constancy of the solubility of the hexagonal germanium oxide GeO_2(hex) over the pH range 1–8 (Baes and Mesmer, 1976; Kossova and Dem'yanets, 1988):

$$GeO_2(hex) + 2H_2O = Ge(OH)_4^\circ(aq) \qquad (K_{hex}) \qquad (2)$$

This result is confirmed by recent Raman spectroscopic (Pokrovski, 1996) and EXAFS (Bernstein and Waychunas, 1987; Pokrovski et al., 1998) measurements. The Raman spectrum of germanic acid shows no changes except intensity with Ge concentration (0.005–0.05m), pH (0.3–7), or temperature (20–250°C). It exhibits a symmetrical band at 710 cm^{-1} which is attributed to the Ge-$(OH)_4$ vibrations. EXAFS measurements conducted at 20°C show that Ge first coordination shell contains four oxygens, and no other Ge atoms are present in the second coordination shell for wide pH (1–10) and concentration (0.002–0.025 m) ranges proving that no polymerisation takes place between the $Ge(OH)_4$ species.

As no changes in the solubility ($pH > 1.5$) or in the Raman spectra of Ge-bearing solutions ($pH > 0.3$) were observed at acid pH, the formation of positively charged Ge hydroxide species can be ruled out at $pH > 0.3$. This is in agreement with the results of GeO_2(hex) solubility measurements performed in the presence of strong acids, HCl, HBr, HNO_3, H_2SO_4, at ambient temperature (Tananaev and Shpirt, 1967). However, these results contradict the UV and extraction measurements of Andrianov and Nazarenko (1966) and Nazarenko and Flyantikova (1968), respectively. These workers proposed the formation of Ge hydroxide cations Ge^{4+}, $Ge(OH)^{3+}$, $Ge(OH)_2^{2+}$, and $Ge(OH)_3^+$ in the pH range 0.2–1.5.

The values of the logarithm of the equilibrium constant of reaction 1 (K_{tetr}) calculated from the average Ge concentrations in equilibrium with GeO_2(tetr) measured in this study in acid and neutral solutions are listed in Table 2 and depicted in Fig. 3 as a function of temperature. It can be seen in this figure that the value of log K_{tetr} determined in this work at 25°C (−5.02) is situated between those of Kossova et al. (1987) (log K_{tetr} = −5.3) and Müller (1926) (log K_{tetr} = −4.4). This value is in excellent agreement with that derived from the solubility product of GeO_2(hex) at 25°C reported in Table 3 (reaction 2) and the Gibbs free energies of formation of GeO_2(tetr) and GeO_2(hex) listed later in Table 5: log K_{tetr}(25°C) = −4.97. At higher temperatures, K_{tetr} values determined in this study are in agreement with those reported by Kossova et al. (1987) from GeO_2(tetr) solubility measurements to 300°C in water.

In basic solutions, GeO_2(tetr) solubility increases with pH (Fig. 4) which is consistent with the dissociation of germanic acid according to the reaction (Baes and Mesmer, 1976)

Table 1. GeO_2(tetr) solubility in the temperature range 25–350°C.

	solution	solid/ solution ratio (R)	pH (25°C)	pH (t°C)T	C_{Ge} (mg/l)	pGe
25°C						
A	HCl-NaCl	1:8.1	4.0	---	0.60	5.08
B	"---"	1:16	4.0	---	0.60	5.08
C	"---"	1:2.1	4.0	---	0.73	5.00
D	"---"	1:1.5	4.0	---	0.75	4.99
E	NaCl	1:3.2	5.5	---	0.50	5.16
50°C						
F	HCl-NaCl	1:2.6	4.0	4.0	2.53	4.46
G	"---"	1:2.6	4.0	4.0	1.34	4.73
H	NaCl	1:6.7	5.5	6.8	2.56	4.45
90°C						
I	HCl-NaCl	1:2.8	4.0	4.1	8.3	3.94
J	"---"	1:3.1	4.5	4.6	5.3	4.14
K	"---"	1:2.6	4.0	4.1	7.9	3.96
L	NaCl	1:3.9	5.5	6.3	9.9	3.86
150°C						
M	$HClO_4$	1:910	1.6	1.7	50	3.16
N	HCl-NaCl	1:33	4.5	4.8	45	3.21
O	"---"	1:27	5.0	5.2	37	3.29
P	HAc-NaAc	1:660	5.1	5.5	30	3.38
Q	"---"	1:730	5.3	5.7	40	3.26
R	NaOH-$B(OH)_3$-NaCl	1:600	8.2	7.8	32	3.35
S	NaOH-$B(OH)_3$	1:540	9.3	8.7	163	2.65
T	"---"	1:615	10.0	9.2	242	2.48
U	NaOH-NaCl	1:275	11.8	9.3	290	2.40
200°C						
V	$HClO_4$	1:400	1.6	1.7	105	2.84
W	HAc-NaAc	1:474	4.7	5.4	145	2.70
X	NaCl	1:36	5.6	5.8	130	2.75
Y	"---"	1:42	5.2	5.5	127	2.76
Z	"---"	1:110	5.2	5.5	105	2.84
250°C						
AA	HCl	1:29	1.9	2.0	432	2.23
AB	"---"	1:29	2.2	2.3	364	2.30
AC	NaCl+H_2O_2	1:10	3.7	3.8	387	2.27
AD	HCl-NaCl + 10 bar O_2	1:100	3.9	4.2	575	2.10
AE	NaCl	1:250	5.5	5.7	510	2.15
AF	H_2O	1:130	6.2	5.6	680	2.03
AG	HCl-NaCl	1:8.5	5.7	5.7	672	2.03
AH	"---"	1:9	6.3	5.5	680	2.03
300°C						
AI	HCl-NaCl	1:130	3.0	3.1	1430	1.71
AJ	H_2O	1:47	7.3	5.7	1300	1.75
AK	"---"	1:300	5.7	5.7	1400	1.71
AL	"---"	1:16	5.1	5.7	880	1.91
AM	"---"	1:8	5.7	5.7	1030	1.85
AN	"---"	1:17	6.4	5.7	1690	1.63
AO	"---"	1:28	5.2	5.7	1270	1.76
350°C						
AP	HCl	1:38	2.0	2.1	1750	1.62
AQ	NaCl	1:3	5.6	5.6	1380	1.72
AR	"---"	1:3	5.7	5.6	1600	1.66
AT	"---"	1:3	5.9	5.6	1520	1.68

pGe stands for the negative logarithm of aqueous Ge molality
Ionic strength ranges between 0 and 0.1 for all runs.

Table 2. Average Ge concentrations in equilibrium with GeO_2(tetr) in acid and neutral solutions, and the logarithms of the equilibrium constant for reaction (1).

T°C	C_{Ge} (mg/l)	log K_{tetr}
25	0.7±0.2	-5.02±0.13
50	2.0±0.7	-4.57±0.20
90	8.0±2.0	-3.96±0.15
150	40±10	-3.26±0.12
200	125±25	-2.76±0.10
250	500±100	-2.16±0.10
300	1200±200	-1.78±0.07
350	1600±100	-1.68±0.05

$$Ge(OH)_4^\circ = GeO(OH)_3^- + H^+ \qquad (K_{dis}) \qquad (3)$$

The value of $Ge(OH)_4^\circ$ dissociation constant (K_{dis}) at 150°C was deduced from the difference between Ge concentrations in equilibrium with GeO_2(tetr) in alkaline (m_{tot}) and acid to neutral solutions ($mGe(OH)_4^\circ$):

$$K_{dis} = (m_{tot} - m(Ge(OH)_4^\circ) \cdot \gamma(GeO(OH)_3^-) \cdot 10^{-pH}/m(Ge(OH)_4^\circ) \quad (4)$$

where $GeO(OH)_3^-$ activity coefficient ($\gamma GeO(OH)_3^-$) was calculated using the Debye-Hückel equation with a° = 4.5Å. K_{dis} value derived at 150°C, $10^{-8.43\pm0.10}$, is in excellent agreement with the corresponding value generated from our potentiometric measurements: $K_{dis} = 10^{-8.41\pm0.05}$ (see below).

3.2. Potentiometric Measurements

The values of pH measured in the two experimental $Ge(OH)_4$-KOH solutions as a function of temperature are listed in Table 4. It can be seen that the pH values determined at the same temperature (50, 75 and 90°C) using a combination pH electrode and a solid-contact pH electrode are the same in the limit of their uncertainties (±0.05 pH). K_{dis} values were calculated for each temperature from the measured pH's using charge-balance and mass-balance constraints for Ge

$$m(H^+) + m(K^+) = m(OH^-) + m(GeO(OH)_3^-), \qquad (5)$$

$$m(Ge_{tot}) = m(Ge(OH)_4^\circ) + m(GeO(OH)_3^-). \qquad (6)$$

In these calculations, the second dissociation product of germanic acid, $GeO_2(OH)_2^{2-}$, was neglected as this species forms in significant amounts only at pH > 12.5 (Baes and Mesmer, 1976; Pokrovski, 1996).

The calculated pK_{dis} values for both experimental solutions are reported in Table 4. It can be seen that the values derived

Table 3. Negative logarithms of the equilibrium constants for reactions (1) and (2) available in the literature.

phase	T°C	$pK_{tetr\ or\ hex}$	reference
GeO_2 hexagonal pK_{hex} reaction (2)	25	1.37	Müller, 1926
	25	1.37	Pugh, 1929
	11	1.42	Schwartz and Huf, 1931
	20	1.39	
	26	1.35	
	35	1.28	
	41	1.23	
	25	1.36	Laubergayer and Morton, 1932
	40	1.22	Evdokimov and Kogan, 1963
	60	1.12	
	80	1.01	
	100	0.94	
	25	1.38	Gayer and Zajicek, 1964
	0	1.59	Vehov et al., 1964
	25	1.32	
	50	1.14	
	75	1.03	
	100	0.94	
GeO_2 tetragonal pK_{tetr} reaction (1)	25	4.37	Müller, 1926
	25	5.34±0.3	Kossova et al., 1987
	50	4.54±0.2	
	100	3.42±0.15	
	160	2.60±?	
	190	2.35±0.2	
	250	1.96±0.1	
	300	1.77±0.07	

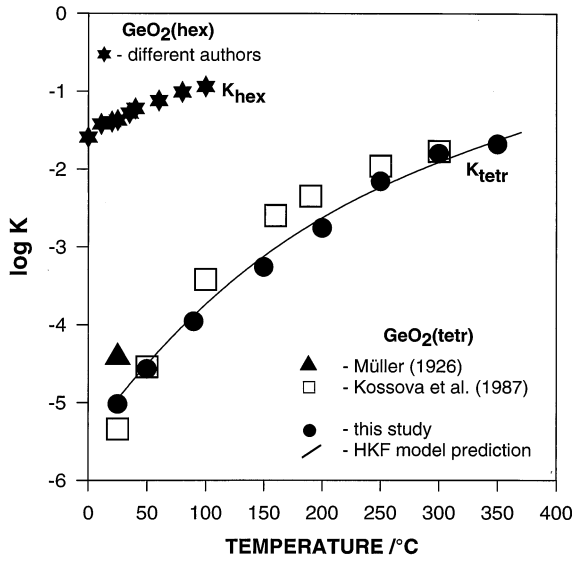


Fig. 3. Logarithms of GeO_2(tetr) and GeO_2(hex) dissociation constants as a function of temperature. The symbols were generated from experiments performed in this study (Table 2) and previous works (Table 3), but the curve was retrieved with the revised HKF model using the parameters given in Tables 5 and 6.

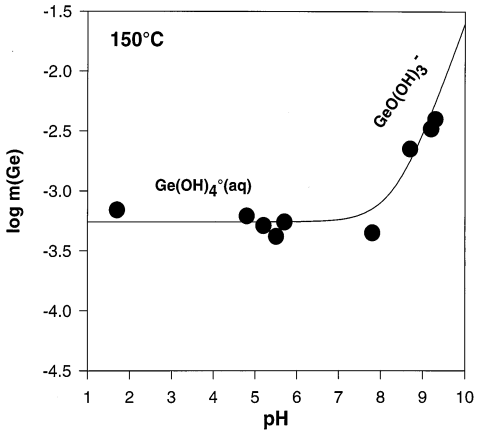


Fig. 4. Logarithm of GeO_2(tetr) solubility as a function of pH at 150°C and Psat. The curve was calculated using values of K_{tetr} and K_{dis} (reactions 1 and 3) generated in this study.

Table 4. pH values measured in $Ge(OH)_4$-KOH solutions, and negative logarithms of the dissociation constant of germanic acid (reaction 3).

t/°C	$Ge(OH)_4$ 0.0080 m KOH 0.0050 m		$Ge(OH)_4$ 0.0184 m KOH 0.0100 m		pK_{dis} average
	pH	pK_{dis}	pH	pK_{dis}	
21*	9.47	9.33	9.45	9.41	9.37
25	---	---	---	---	**9.32±0.05**
31*	9.38	9.23	9.31	9.28	9.26
50*	9.11	8.94	9.03	9.01	---
50	9.02	8.84	9.02	9.00	8.92
60*	9.01	8.84	---	---	8.84
75*	8.85	8.69	---	---	---
75	8.79	8.63	8.77	8.77	8.70
90*	8.77	8.64	---	---	---
90	8.71	8.58	8.68	8.67	8.63
100	8.63	8.50	8.59	8.59	8.55
125	8.53	8.45	8.49	8.50	8.48
150	8.42	8.40	8.38	8.42	8.41
175	8.31	8.32	8.31	8.38	8.35
190	8.27	8.32	8.27	8.35	8.34
200	8.25	8.30	8.32	8.43	8.37

* - measurements performed with a combination pH electrode.

at the same temperature for each solution are the same in the limit of the calculation uncertainties ($< \pm 0.08$ pK). Thus, the average pK_{dis} values were calculated from these two series of measurements. They are listed in Table 4 and plotted as a function of temperature in Fig. 5. The value at 25°C was retrieved using a linear interpolation of pK_{dis} values measured at 21, 31, and 50°C as a function of the reciprocal absolute temperature. This value ($pK_{dis} = 9.32 \pm 0.05$) is in excellent agreement with that of Baes and Mesmer (1976) ($pK_{dis} = 9.3 \pm 0.2$) which is based on the potentiometric measurements of Ingri (1963) and Haas et al. (1964). In contrast, the pK_{dis} values determined by Antikainen (1957) in the temperature range 10–45°C from potentiometric titrations of 0.03–0.04 m germanic acid solutions are considerably lower ($\Delta pK > 0.6$) than the corresponding values obtained in this study (Fig. 5). Furthermore, the value of the enthalpy of reaction 3 derived by Antikainen (1957) from his pK values ($\Delta H° = 12$ kcal/mol at 25°C) is considerably higher than those determined in this work ($\Delta H° = 6.5$ kcal/mol) and in previous studies on dissociation of the analogous arsenious, boric and silicic acids ($\Delta H° = 3$–7 kcal/mol, Baes and Mesmer, 1976; Mesmer et al., 1972; and Busey and Mesmer, 1977). This discrepancy is likely due to the formation of polymeric germanate species in the concentrated Ge solutions ($m_{Ge} > 0.03$ m) used by Antikainen (1957).

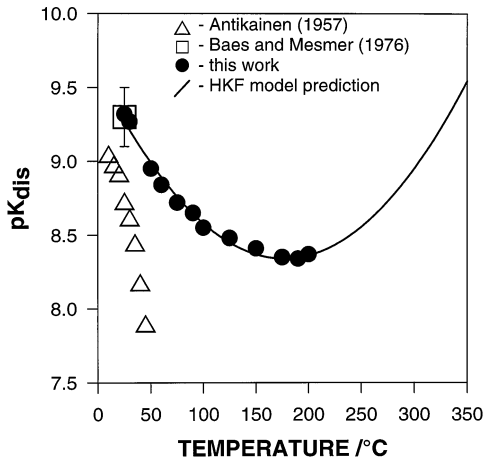


Fig. 5. Negative logarithm of germanic acid dissociation constant (reaction 3) as a function of temperature at P_{sat}. The symbols depict experimental data generated in this work and taken from the literature, but the curve was retrieved with the revised HKF model using the parameters given in Tables 5 and 6.

4. DETERMINATION OF THE THERMODYNAMIC PROPERTIES OF GERMANIUM HYDROXIDE COMPLEXES

4.1. Conventions, Units and Standard States

The standard molal Gibbs free energies ($\Delta_f G°$) and enthalpies ($\Delta_f H°$) of compounds adopted in this study represent the change in those thermodynamic quantities when one mole of the substance in its standard state is formed isothermally from the elements at 1 bar pressure. The standard states for the solid phases and H_2O are unit activity for the pure phase at all temperatures and pressures. For aqueous species, the reference state convention corresponds to unit activity coefficient for an hypothetical one molal solution whose behaviour is ideal.

4.2. Thermodynamic Properties of Germanium Oxides

The generation of thermodynamic properties for aqueous Ge species from solubility data requires corresponding properties for the solid Ge phases. A detailed review of the available thermodynamic data for hexagonal and tetragonal germanium oxides are given elsewhere (Pokrovski, 1996). The standard molal entropies ($S°_{298}$) of GeO_2(hex) and GeO_2(tetr) were retrieved from low temperature heat capacity measurements of King (1958) and Counseil and Martin (1967), respectively. Equations describing the heat capacities ($C°_p$) of these phases as a function of temperature above 25°C were generated by Pankratz (1982) from high temperature enthalpy and heat capacity measurements of Andon and Mills (1971), Tsagareyshvili and Gvelesiani (1973), Kelly and Cristensen (1961), and Gant (1967). The standard molal enthalpies of formation ($\Delta_f H°_{298}$) of GeO_2(hex) and GeO_2(tetr) were taken from fluorine combustion calorimetric measurements reported by Gross et al. (1966). The standard molal Gibbs free energies of formation at 25°C ($\Delta_f G°_{298}$) were calculated from $\Delta_f H°_{298}$ and $S°_{298}$ of GeO_2(hex) and GeO_2(tetr), and the entropies of Ge(cr) and O_2(g) taken from Pankratz (1982). The Gibbs free energies, enthalpies, and entropies adopted in this study are in excellent agreement with those reported in the thermodynamic handbooks (Pankratz, 1982; Glushko, 1979a; Wagman et al., 1982). The standard molar volume (V°) of GeO_2(hex) was taken from Robie et al. (1978), while that of GeO_2(tetr) was calculated from the unit-cell parameters of the oxide. The thermodynamic parameters adopted in this study for germanium oxides are listed in Table 5.

4.3. Thermodynamic Properties of $Ge(OH)°_4$(aq) and $GeO(OH)^-_3$

The thermodynamic parameters of $Ge(OH)°_4$(aq) were generated within the framework of the HKF model (Helgeson et

Table 5. Thermodynamic properties of GeO_2(tetr) et GeO_2(hex) used inthis study.

	$\Delta_f G°_{298}$ kcal/mol	$\Delta_f H°_{298}$ kcal/mol	$S°_{298}$ cal/mol.K	$C°_{p\,298}$ cal/mol.K	$V°_{298}$ cm^3/mol
GeO_2(hex)	-119.7±0.3 (4)	-132.6±0.3 (3)	13.21±0.10 (1)	12.45 (5)	24.44 (7)
GeO_2(tetr)	-124.6±0.7 (4)	-138.7±0.7 (3)	9.49±0.04 (2)	11.99 (6)	17.42 (8)

(1) King (1958); (2) Counseil and Martin (1967)

(3) Gross et al. (1966)

(4) calculated from (1) and (3), and (2) and (3) using S° of Ge(cr) and O_2(g) from Pankratz (1982)

(5) $C°_p = 14.623 + 3.296\cdot10^{-3}T - 2.805\cdot10^{+5}/T^2$ (25-1116°C) (Pankratz, 1982)

(6) $C°_p = 16.457 + 2.178\cdot10^{-3}T - 4.549\cdot10^{+5}/T^2$ (25-1035°C) (Pankratz, 1982)

(7) Robie et al. (1978)

(8) calculed from the unit-cell parameters of GeO_2(tetr).

al., 1981; Tanger and Helgeson, 1988) using values of K_{tetr} obtained in this study and those of K_{hex} available in the literature. Data regression was performed using HELGESON (Akinfiev, 1993) and UT-HEL (Shvarov, 1993) computer codes which are consistent with the revised HKF equation of state (Tanger and Helgeson, 1988). The apparent molal Gibbs free energies ($G°_T$) of $Ge(OH)^\circ_4$(aq) were calculated using GeO_2(tetr) and GeO_2(hex) thermodynamic parameters given in Table 5. Those of H_2O are taken from SUPCRT92 (Johnson et al., 1992) and equilibrium constants for reactions 1 and 2 from Tables 2 and 3, respectively. $C°_p$, and c_1 and c_2 HKF coefficients were generated by fitting $Ge(OH)^\circ_4$(aq) Gibbs free energy values from 20 to 350°C. This regression was performed by adopting $\Delta_f G°_{298}$ and $S°_{298}$ values for $Ge(OH)^\circ_4$(aq) listed in Table 6. $\Delta_f G°_{298}$ was calculated from pK_{hex}(25°C) and $\Delta_f G°_{298}$ (GeO_2(hex)). The value of $S°_{298}$ was retrieved from the variation of pK_{hex} and pK_{tetr} as a function of reciprocal temperature from 20 to 60°C. The standard partial molal volume of $Ge(OH)^\circ_4$(aq) at 25°C (54.9 cm^3/mol) was derived from $V°Si(OH)^\circ_4$(aq) (52.3 cm^3/mol, SUPCRT92) assuming that the difference between $V°Si(OH)^\circ_4$(aq) and $V°Ge(OH)^\circ_4$(aq) is the same as that between $V°As(OH)^\circ_3$(aq) (49 cm^3/mol; Pokrovski et al., 1996a) and $V°Sb(OH)^\circ_3$(aq) (51.6 cm^3/mol; Shikina and Zotov, 1990). a_1-a_4 HKF coefficients were generated from $V°Ge(OH)^\circ_4$(aq) value together with correlations reported by Shock et al. (1989) for neutral aqueous species. In the absence of direct high temperature $Ge(OH)^\circ_4$(aq) heat capacity and volume measurements, the Born parameter (ω) was assigned a value of zero. This value is in agreement with those adopted for other neutral species (e.g., $B(OH)^\circ_3$(aq), $NaB(OH)^\circ_4$(aq), Pokrovski et al., 1995; $As(OH)^\circ_3$(aq), Pokrovski et al., 1996a; $Al(OH)^\circ_3$(aq), and $NaAl(OH)^\circ_4$(aq), Diakonov et al., 1996). The HKF state parameters of $Ge(OH)^\circ_4$(aq) generated in this study are reported in Table 6. The values of log K_{tetr} calculated using these parameters are shown by the solid line in Fig. 3.

The thermodynamic parameters of $GeO(OH)^-_3$ were generated using the values of K_{dis} obtained in this study. The apparent Gibbs free energies of $GeO(OH)^-_3$ as a function of temperature were calculated from those for $Ge(OH)^\circ_4$(aq) derived above and pK_{dis} determined in this work. The value of $C°_p(GeO(OH)^-_3)$ was derived from $C°_p(Ge(OH)^\circ_4)$ and the $\Delta C_p(298)$ of reaction proposed by Smith et al. (1986) for the dissociation reactions of weak oxy-acids:

$$H_nMO_x = H^+ + H_{n-1}MO^-_x \qquad (\Delta C_{p,298} = -32.0 \text{ cal/(mol.K)}).$$

The value of $S°_{298}(GeO(OH)^-_3)$ was generated by fitting the apparent Gibbs free energies of $GeO(OH)^-_3$ in the temperature range 21–200°C. V°, $c_{1\text{-}2}$, $a_{1\text{-}4}$ and ω coefficients were retrieved from S° and $C°_p$ values using the correlations reported by Shock and Helgeson (1988) for ions. The HKF parameters of $GeO(OH)^-_3$ are reported in Table 6. The pK_{diss} values generated as a function of temperature using these parameters are shown by the solid line in Fig. 5. It can be seen that these values decrease with increasing temperature, exhibit a minimum near 180°C, and increase at higher temperature. This behavior is identical to that observed for other weak acids like boric (Mesmer et al., 1972), silicic (Busey and Mesmer, 1977), and arsenious (Pokrovski, 1996) acids.

5. COMPARISON OF THE AQUEOUS SPECIATION OF GERMANIUM AND SILICON: IMPLICATION FOR GE/SI RATIOS IN CRUSTAL FLUIDS

Thermodynamic parameters for $Ge(OH)^\circ_4$(aq) and $GeO(OH)^-_3$ retrieved in this study allow to compare the stability fields of these species as a function of pH and temperature with those of the analogous silicon hydroxide species. The distribution of Ge and Si hydroxide complexes are shown as a function of pH at 25 and 250°C in Fig. 6. These diagrams were constructed using the dissociation constants for germanic and silicic acids generated in the present study and taken from Busey and Mesmer (1977), respectively. It can be seen in this figure that the neutral hydroxide species are dominant for both Ge and Si in acid and neutral solutions. The negatively charged hydroxide complexes develop in significant amounts (>20%) only at pH higher than 9–10 at 25°C, and 8–9 at 250°C. The stability field of $GeO(OH)^-_3$ is shifted by ~0.5 pH unit towards acid pH's in comparison with that of $SiO(OH)^-_3$ at the same temperature. This earlier dissociation of germanic acid can increase Ge mobility relative to Si mobility in alkaline solutions. This probably explains the increase of Ge concentrations and Ge/Si ratios at pH > 7–8 observed in the thermal waters from the Pamir and Greater Caucasus (Kraynov, 1967) and in the hydrothermal springs of New Zealand (Koga, 1967).

In most surficial and hydrothermal fluids whose pH values are close to the neutrality point of water, Ge and Si are present in the form of the neutral hydroxides $Ge(OH)^\circ_4$(aq) and $Si(OH)^\circ_4$(aq), respectively (Fig. 6). However, the thermodynamic properties of these species are significantly different. This is the case for the values of $\Delta_f H°_{298}$ and $C°_{p,298}$ which are

Table 6. Standard molal thermodynamic properties at 25°C and 1 bar and HKF equation-of-state parameters for $Ge(OH)_4^\circ(aq)$ and $GeO(OH)_3^-$ retrieved in this study.

$Ge(OH)_4°(aq)$	
$\Delta_fG°$ = -231.2±0.4 kcal/mol (1)	$\Delta_fH°$ = -267±0.9 kcal/mol (3)
$S°$ = 47.83±3.0 cal/(mol K) (2)	$C_p°$ = 50.0±5.0 cal(mol K) (2)
$V°$ = 54.9 cm^3/mol (4)	ω = 0
c_1 = 57.1±4.0 cal/(mol K) (2)	$c_2 \times 10^{-4}$ = -3.5±1.0 cal K/mol (2)
$a_1 \times 10$ = 9.29 cal/(mol bar)	$a_2 \times 10^{-2}$ = 14.9 cal/mol
a_3 = -0.11 cal K/(mol bar)	$a_4 \times 10^{-4}$ = -3.4 cal K/mol

(1) from $\Delta_fG°(GeO_{2,hex})$ and pK_{hex} at 25°C, (2) from the regression of apparent molal Gibbs free energies of $Ge(OH)_4°(aq)$ calculated from pK_{tetr} in the interval 25-350°C, (3) calculated from (1) and (2) using the entropies of $H_2(g)$, $O_2(g)$ and Ge(cr) from Pankratz (1982), (4) from $V°(Si(OH)_4°)$ using the difference between the partial molal volumes of $As(OH)_3°$ and $Sb(OH)_3°$ (see text). a_{1-4} coefficients were found from this value using the HKF correlations for neutral species from Shock et al. (1989).

$GeO(OH)_3^-$	
$\Delta_fG°$ = -218.5±0.60 kcal/mol (1)	$\Delta_fH°$ = -260.5±0.8 kcal/mol (4)
$S°$ = 27.14 cal/(mol K) (2)	$C_p°$ = 18.0 cal(mol K) (5)
$V°$ = 27.14 cm^3/mol (3)	$\omega = 1.20 \times 10^5$ cal/(mol) (6)
c_1 = 27.66 cal/(mol K)	$c_2 \times 10^{-4}$ = 0.63 cal K/mol
$a_1 \times 10$ =5.96 cal/(mol bar)	$a_2 \times 10^{-2}$ = 6.77 cal/mol
a_3 = 3.09 cal K/(mol bar)	$a_4 \times 10^{-4}$ = -3.06 cal K/mol

(1) from $\Delta_fG°(Ge(OH)_4°)$ and pK_{dis} at 25°C, (2) from the regression of apparent molal Gibbs free energies of $GeO(OH)_3^-$ calculated from pK_{dis} in the interval 25-200°C, (3) from the correlation S°-V° for monovalent polyatomic anions (Shock and Helgeson, 1988), (4) from (1) et (2) using the entropies of $H_2(g)$, $O_2(g)$ and Ge(cr) from Pankratz (1982), (5) derived from ΔC_p = -32 cal/(mol K) for dissociation reactions of the type $H_nMO_x = H^+ + H_{n-1}MO_x^-$ (Smith et al., 1986), (6) from electrostatic radius and entropy of $GeO(OH)_3^-$ (Shock and Helgeson, 1988). $c_{1,2}$ and a_{1-4} coefficients were determined from Cp° and V°, respectively, using the correlations for ions from Shock and Helgeson (1988).

equal to −346.2 kcal/mol and −40 cal/mol.K (SUPCRT92), and −267.0 and +50 cal/mol.K (this study) for $Si(OH)_4^\circ(aq)$ and $Ge(OH)_4^\circ(aq)$, respectively. This difference can lead to higher values for the $\Delta H°$ and ΔC_p° of the dissociation reactions of germanates than silicates and thus increase Ge/Si ratio with temperature in a solution in equilibrium with a Ge-bearing silicate (where Si is partially substituted by Ge). To illustrate this possible effect, the $Ge(OH)_4^\circ(aq)/Si(OH)_4^\circ(aq)$ molal ratio in an aqueous solution in equilibrium with a Ge-bearing wollastonite ($Ca(Si_{1-x}Ge_x)O_3$) has been calculated as a function of temperature. Equilibrium of an aqueous solution with the two end-members of the Si-Ge solid solution

$$CaSiO_3 + 2H^+ + H_2O = Si(OH)_4^\circ(aq) + Ca^{2+} \quad K_{CaSiO_3} \quad (7)$$

$$CaGeO_3 + 2H^+ + H_2O = Ge(OH)_4^\circ(aq) + Ca^{2+} \quad K_{CaGeO_3} \quad (8)$$

yields

$$\frac{a_{Ge(OH)_4^\circ}}{a_{Si(OH)_4^\circ}} = \frac{K_{CaGeO_3}}{K_{CaSiO_3}} \times \frac{a_{CaGeO_3}}{a_{CaSiO_3}}, \quad (9)$$

where K_{CaGeO3} and K_{CaSiO3} stand for the dissociation constants of $CaGeO_3$ and $CaSiO_3$, respectively, a_{CaGeO_3} and a_{CaSiO_3} denote the activities of $CaGeO_3$ and $CaSiO_3$ in the solid solution, respectively. $a_{Ge(OH)_4°}$ and $a_{Si(OH)_4°}$ represent the activities of germanic and silicic acid in the aqueous solution and can be replaced by their corresponding molalities as the activity coefficients of neutral species are close to 1 in dilute solutions. Figure 7 shows a plot of the log K_{CaGeO_3}, log K_{CaSiO_3} and log $(mGe(OH)_4^\circ/mSi(OH)_4^\circ)$ in an aqueous solution in equilibrium

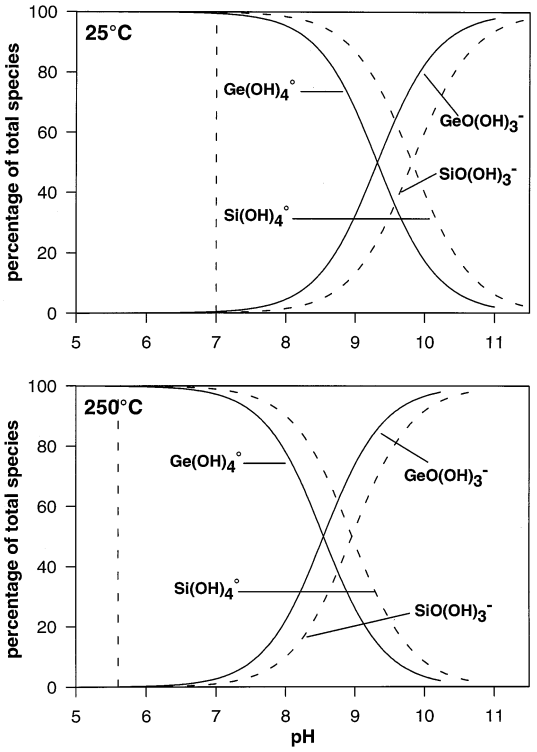


Fig. 6. Distribution of Ge and Si hydroxide species as a function of pH at 25 and 250°C, and P_{sat}. The distribution curves were calculated using the parameters given in Table 6 and in SUPCRT92 for Ge and Si aqueous hydroxide species, respectively. The vertical dashed lines stand for the pH of the neutrality point of water at a given temperature.

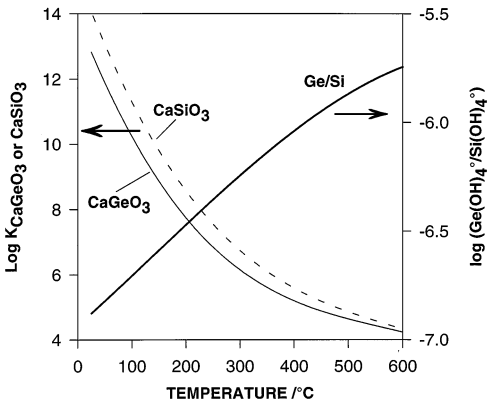


Fig. 7. Log K of $CaSiO_3$ (wollastonite) and $CaGeO_3$ (Eqns. 7 and 8), and of $Ge(OH)_4^\circ/Si(OH)_4^\circ$ ratio in a solution in equilibrium with a germanium-bearing wollastonite containing 1.4 ppm Ge as a function of temperature and at 1 kbar.

with a wollastonite containing 1.4 ppm of Ge ($x = 2.2 \cdot 10^{-6}$) as a function of temperature at P = 1 kbar. To perform this calculation, it was assumed that the solid solution $Ca(Si_{1-x}Ge_x)O_3$ behaves ideally for minor amounts of Ge in the solid phase ($a_{CaGeO_3} = x$ and $a_{CaSiO3} = 1-x$). K_{CaGeO_3} and K_{CaSiO_3} values were calculated using the thermodynamic parameters of $CaGeO_3$ by Glushko (1979b), $Ge(OH)_4^\circ$(aq) generated in this study, and those of wollastonite, $Si(OH)_4^\circ$(aq), Ca^{2+}, and H_2O taken from SUPCRT92 (Johnson et al., 1992). It can be seen in Fig. 7 that the $Ge(OH)_4^\circ/Si(OH)_4^\circ$ ratio increases by more than an order of magnitude when the temperature is raised from 25 to 500°C. This calculated increase of the Ge/Si ratio with temperature is of the same order of magnitude than that observed in natural hydrothermal fluids. For example, in Fig. 8 is plotted the logarithm of Ge/Si ratios as a function of temperature, deduced from Ge and Si concentrations reported by Arnorsson (1984) in Icelandic hydrothermal waters. The average Ge/Si ratio depicted by the solid line that represents a linear regression of all available data increases by a half

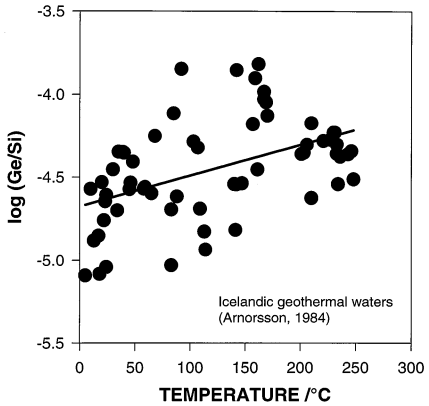


Fig. 8. Ge/Si atomic ratio in Icelandic geothermal well waters as a function of temperature, calculated from Ge and Si concentrations reported by Arnorsson (1984). The symbols stand for samples from different locations, but the solid line represents a linear regression of all data.

an order of magnitude when temperature rises from 25 to 250°C. This is consistent with Ge/Si increase predicted in Fig. 7. The difference in the thermodynamic properties of $Ge(OH)_4^\circ$(aq) and $Si(OH)_4^\circ$(aq) could thus be responsible for the increase of Ge/Si ratios measured in crustal fluids at high temperature.

6. CONCLUDING REMARKS

The stability of Ge(IV) hydroxide species were determined from (a) solubility measurements of GeO_2(tetr) in acid to basic solutions to 350°C, and (b) pH measurements on germanic acid-KOH aqueous solutions to 200°C. The experimental data obtained in this study were combined with the thermodynamic properties of Ge (IV) oxides available in the literature to generate, within the framework of the revised HKF model, the standard partial molal properties and equations of state parameters for $Ge(OH)_4^\circ$(aq) and $GeO(OH)_3^-$.

Species distribution calculations performed using these parameters indicate that $Ge(OH)_4^\circ$(aq) dominates Ge speciation in acid to neutral solutions (pH < 8) at temperatures up to at least 350°C. $GeO(OH)_3^-$ species can occur in significant amounts (>20%) only in alkaline solutions (pH > 8–9). This speciation distribution is very close to that of aqueous silica.

The difference between the thermodynamic properties of $Ge(OH)_4^\circ$(aq) and $Si(OH)_4^\circ$(aq) leads to the increase of the Ge/Si ratio with temperature in aqueous solutions in equilibrium with Ge-bearing silicates. This can account for the high values of Ge/Si measured in high temperature crustal fluids.

In low temperature surficial waters the Ge/Si ratio can be affected by the changes in chemical-weathering intensity, by different chemical composition of the rocks exposed to alteration, and by the presence of dissolved organic matter. The influence of aqueous organic acids which can strongly complex Ge vs. Si on the aqueous Ge/Si ratios in superficial waters is discussed in our subsequent paper (Pokrovski and Schott, 1998).

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